

Unit ② P ③

Important questions:

- ✓ ① Suppose that X and Y are independent normalized normal random variables. Find the pdf of

$$Z = X + Y$$

Ans: $f_z(z) = \frac{1}{\sqrt{2\pi}\sqrt{2}} e^{-z^2/(2(\sqrt{2})^2)}$ or $N(0, \sqrt{2})$

- ✓ ② Let $Z = aX + bY$ where ' a ' and ' b ' are arbitrary constants. Show that if X and Y are independent, then

$$\sigma_z^2 = a^2 \sigma_x^2 + b^2 \sigma_y^2$$

- ✓ ③ Let ' X ' and ' Y ' be defined by:

$$X = \cos \phi \quad \text{and} \quad Y = \sin \phi$$

where ϕ is a random variable uniformly distributed over $[0, 2\pi]$

(a) Show that ' X ' and ' Y ' are uncorrelated

(b) Show that ' X ' and ' Y ' are not independent

- ✓ ④ Let (X, Y) be a bivariate r.v. with the joint pdf,

$$f_{XY}(x, y) = \frac{x^2 + y^2}{4\pi} e^{-(x^2 + y^2)/2}$$

where $-\infty < x, y < \infty$

Show that X and Y are not independent
~~but are uncorrelated~~

⑤ Let X and Y be two independent random variables with

$$f_X(x) = \alpha e^{-\alpha x} u(x)$$

$$f_Y(y) = \beta e^{-\beta y} u(y)$$

Find the density function of $Z = X + Y$

⑥ Find the moment generating function of 'X' uniformly distributed over (a, b) .

(b) Using the result of (a), find $E[X]$, $E[X^2]$ and $E[X^3]$.

⑦ Find the covariance of 'X' and 'Y' if
(a) they are independent (b) Y is related to 'X' by $Y = ax + b$

Ans: (a) $\sigma_{XY} = 0$ and (b) $\sigma_{XY} = C_{XY} = a\sigma_X^2$

⑧ Let 'X' and Y be two jointly continuous random variables with joint pdf.

$$f_{XY}(x, y) = \begin{cases} 2 & x+y \leq 1, x > 0, y > 0 \\ 0 & \text{o.w.} \end{cases}$$

find (i) $\text{cov}(X, Y)$

(ii) $\rho(X, Y)$

9 ✓ Let 'X' and 'Y' be jointly continuous random variables with joint pdf.

$$f_{XY}(u, v) = \begin{cases} cu + 1 & u, v \geq 0 \quad u+v < 1 \\ 0 & \text{o.w.} \end{cases}$$

Find

- (i) constant 'c'
- (ii) Marginal PDF's $f_X(u)$ & $f_Y(v)$
- (iii) $P[X < 2X^2]$

10 ✓ Let 'X' be a continuous random variable with pdf.

$$f_X(u) = \begin{cases} 4u^3 & 0 \leq u \leq 1 \\ 0 & \text{o.w.} \end{cases}$$

Let $Y = 1/X$ then Find $f_Y(v)$

Solu: Hint: $f_Y(v) = \frac{f_X(u)}{\left| \frac{dy}{du} \right|} = \frac{f_X(u)}{\left| -\frac{1}{u^2} \right|}$

Ans: $f_Y(v) = \begin{cases} 4/v^5 & v \geq 1 \\ 0 & \text{o.w.} \end{cases}$

①. Let 'x' be a continuous random variable with pdf

$$f_x(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \text{ for all 'x'}$$

Let $Y = X^2$. Find $f_Y(y)$

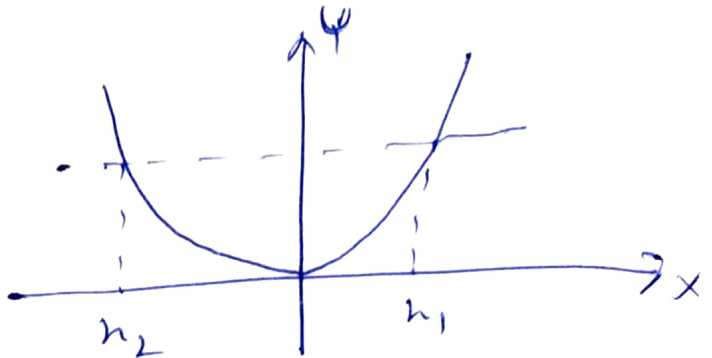
Solution: Hint:

Since

$$Y = X^2$$

$$\Rightarrow x = \pm \sqrt{y}$$

$$\Rightarrow x_1 = \sqrt{y}, x_2 = -\sqrt{y}$$



$$\Rightarrow f_Y(y) = \frac{f_x(x_1)}{\left| \frac{d f_x(x)}{dx} \right|_{x=x_1}} + \frac{f_x(x_2)}{\left| \frac{d f_x(x)}{dx} \right|_{x=x_2}}$$

$$\Rightarrow f_Y(y) = \frac{f_x(\sqrt{y})}{|2\sqrt{y}|} + \frac{f_x(-\sqrt{y})}{|-2\sqrt{y}|}$$

$$\Rightarrow f_Y(y) = \frac{1}{2\sqrt{2\pi y}} e^{-y/2} + \frac{1}{2\sqrt{2\pi y}} e^{-y/2}$$

$$\Rightarrow f_Y(y) = \frac{1}{\sqrt{2\pi y}} e^{-y/2}$$

Ans

12 Let 'X' be a continuous random variable with pdf given by

$$f_X(u) = \frac{1}{2} e^{-|u|}$$

if $Y = X^2$, find the CDF of Y.

Hint:

$$F_Y(y) = P(Y \leq y)$$

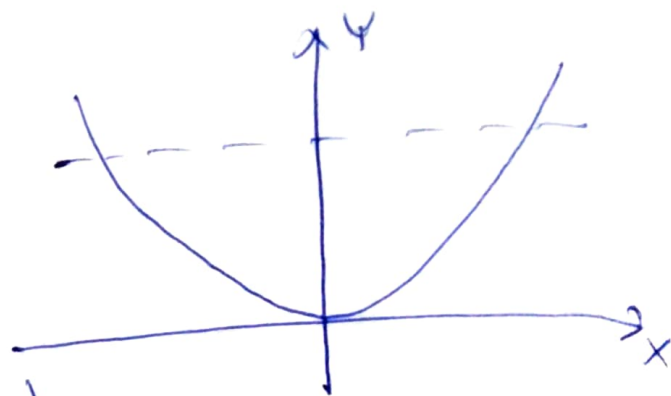
$$= P(X^2 \leq y)$$

$$= P(-\sqrt{y} \leq X \leq \sqrt{y})$$

$$= \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2} e^{-|u|} du$$

$$= \int_0^{\sqrt{y}} e^{-|u|} du = 1 - e^{-\sqrt{y}}$$

$$\Rightarrow F_Y(y) = \begin{cases} 1 - e^{-\sqrt{y}} & y \geq 0 \\ 0 & \text{o.w} \end{cases}$$



13 Let 'X' be a continuous random variable with pdf

$$f_X(u) = \begin{cases} 4u^3 & 0 \leq u \leq 1 \\ 0 & \text{o.w} \end{cases}$$

Find $P(X \leq 2/3 \mid X > 1/3)$

Ans: $3/16$

PTSP Assignment-2 (Unit @ 2 and 3)

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[Because when

we sum the P.V.s

the resultant

is convolution of

two]

Q1

$$z = x + y$$

$$f_z(z) = \int_{-\infty}^{\infty} f_x(x) f_y(z-x) dx$$

$$f_z(z) = \int_{-\infty}^{\infty} \frac{1}{(\sqrt{2\pi})\sigma_x\sigma_y} e^{-\frac{1}{2}\left(\frac{x-\mu_x}{\sigma_x}\right)^2} \cdot e^{-\frac{1}{2}\left(\frac{z-x-\mu_y}{\sigma_y}\right)^2} dx$$

Consider $\mu_x = \mu_y = 0$ and $\sigma_x = \sigma_y = 1$

$$f_z(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{1}{2}[x^2 + (z-x)^2]} dx$$

$$f_z(z) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(2x^2 + z^2 - 2zx)} dx$$

$$f_z(z) = \frac{1}{2\pi} e^{-z^2/2} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x^2 - zx)} dx$$

$$f_z(z) = \frac{1}{2\pi} e^{-z^2/2} \cdot e^{z^2/4} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x^2 - zx + z^2/4)} dx$$

$$f_z(z) = \frac{1}{2\pi} e^{-z^2/4} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(x - z/2)^2} dx$$

WKT, $1 = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}x^2} dx$ $\left[\int_{-\infty}^{\infty} f_x(x) dx = 1 \right]$

$$\int_{-\infty}^{\infty} e^{-x^2/2} dx = \sqrt{2\pi}$$

$$f_z(z) = \frac{1}{2\pi} e^{-z^2/4} \frac{1}{\sqrt{2}} \int_{-\infty}^{\infty} e^{-\frac{1}{2} \left(x - z\sqrt{2}/2\right)^2} dx$$

$$f_z(z) = \frac{1}{\sqrt{2\pi}(\sqrt{2})} e^{-z^2/2(\sqrt{2})^2}$$

$$\text{OR } f_z(z) = N(0, \sqrt{2})$$

Q2

$$Z = aX + bY$$

$$E(Z) = aE(X) + bE(Y)$$

$$Z^2 = a^2X^2 + b^2Y^2 + 2abXY$$

$$E(Z^2) = a^2E(X^2) + b^2E(Y^2) + 2abE(XY)$$

If X and Y are independent $E(XY) = E(X)E(Y)$

$$E(Z^2) = a^2E(X^2) + b^2E(Y^2) + 2abE(X)E(Y)$$

$$\sigma_Z^2 = E(Z^2) - [E(Z)]^2$$

$$\sigma_Z^2 = a^2E(X^2) + b^2E(Y^2) + 2abE(X)E(Y) - [a^2E(X)^2 + b^2E(Y)^2 + 2abE(X)E(Y)]$$

$$\sigma_Z^2 = a^2[E(X^2) - (E(X))^2] + b^2[E(Y^2) - (E(Y))^2]$$

$$\sigma_Z^2 = a^2\sigma_X^2 + b^2\sigma_Y^2$$

Q3

$$X = \cos \phi \text{ and } Y = \sin \phi$$

$$\phi \sim U[0, 2\pi]$$

$$(a) E(X) = \int_0^{2\pi} \cos \phi \frac{1}{2\pi} d\phi = 0$$

$$E(Y) = \int_0^{2\pi} \sin \phi \frac{1}{2\pi} d\phi = 0$$

$$E(XY) = \int_0^{2\pi} 8 \sin \phi \cos \phi \frac{1}{2\pi} d\phi$$

$$= \frac{1}{4\pi} \int_0^{2\pi} 8 \sin 2\phi d\phi$$

$$E(XY) = 0$$

as $E(XY) = E(X)E(Y) = 0$

$\Rightarrow C_{XY} = 0 \Rightarrow \text{Uncorrelated}$

(b) for independent $E(XY) = E(X)E(Y)$ and also
 $E(X^2Y^2) = E(X^2)E(Y^2)$

$$E(X^n Y^n) = E(X^n)E(Y^n)$$

$$E(X^2) = \frac{1}{2\pi} \int_0^{2\pi} \cos^2 \phi d\phi = \frac{1}{2} \quad \left. \begin{array}{l} \\ \\ \end{array} \right\} E(X^2)E(Y^2) = \frac{1}{4}$$

$$E(Y^2) = \frac{1}{2\pi} \int_0^{2\pi} \sin^2 \phi d\phi = \frac{1}{2}$$

$$E(X^2Y^2) = \frac{1}{2\pi} \int_0^{2\pi} \sin^2 \phi \cos^2 \phi d\phi$$

$$= \frac{1}{2\pi} \times \frac{1}{4} \int_0^{2\pi} \sin^2 2\phi d\phi = \frac{1}{16\pi} \int_0^{2\pi} (1 - \cos 4\phi) d\phi$$

$$= \frac{1}{8}$$

We can see $E(X^2Y^2) \neq E(X^2)E(Y^2)$

This means X and Y are not independent.

Q4

$$f_{xy}(x, y) = \frac{x^2 + y^2}{4\pi} e^{-\frac{(x^2 + y^2)}{2}} \quad -\infty < x, y < \infty$$

$$f_x(x) = \int_{-\infty}^{\infty} \frac{x^2 + y^2}{4\pi} e^{-\frac{x^2}{2}} \cdot e^{-\frac{y^2}{2}} dy$$

$$= \frac{e^{-x^2/2}}{4\pi} \left[x^2 \int_{-\infty}^{\infty} e^{-y^2/2} dy + 2 \int_{-\infty}^{\infty} y \cdot \frac{y}{2} e^{-y^2/2} dy \right]$$

$$= \frac{e^{-x^2/2}}{4\pi} \left[\sqrt{2\pi} x^2 + 2 \left[-y e^{-y^2/2} \right]_0^{\infty} + 2 \int_0^{\infty} e^{-y^2/2} dy \right]$$

$$f_x(x) = \frac{e^{-x^2/2} \sqrt{2\pi}}{4\pi} [x^2 + 1]$$

Similarly

$$f_y(y) = \frac{e^{-y^2/2} \sqrt{2\pi}}{4\pi} [y^2 + 1]$$

$$f_x(x) \cdot f_y(y) = \frac{e^{-(x^2 + y^2)/2}}{8\pi} [x^2 + 1] [y^2 + 1]$$

$$\neq f_{xy}(x, y)$$

$\Rightarrow$ x and y are not independent

Q5 $f_x(x) = \alpha e^{-\alpha x} u(x) \quad f_y(y) = \beta e^{-\beta y} u(y)$

$$z = x + y$$

$$f_z(z) = \int_{-\infty}^{\infty} f_x(x) f_y(z-x) dx$$

$$= \int_{-\infty}^{\infty} \alpha e^{-\alpha x} u(x) \beta e^{-\beta(z-x)} u(z-x) dx$$

$$= \alpha \beta \int_0^z e^{-\alpha x} \cdot e^{-\beta(z-x)} dx$$

$$f_x(z) = \alpha \beta e^{-\beta z} \int_a^z e^{(\beta-\alpha)u} du$$

$$f_x(z) = \frac{\alpha \beta e^{-\beta z}}{(\beta-\alpha)} [e^{(\beta-\alpha)z} - 1]$$

Q6

$$X \sim U[a, b]$$

$$\begin{aligned} \text{(a)} \quad M_X(u) &= E(e^{ux}) = \int_a^b e^{ux} f_X(x) dx \\ &= \int_a^b e^{ux} \frac{1}{b-a} dx \end{aligned}$$

$$M_X(u) = \frac{1}{(b-a)u} (e^{ub} - e^{ua})$$

Q6

$$X \sim U[a, b]$$

$$(b) \quad E(X) = \left. \frac{dM_X(u)}{du} \right|_{u=0}$$

$$M_X(u) = \frac{1}{(b-a)u} \left[e^{bu} - e^{au} \right]$$

WKT

$$e^{bu} = 1 + \frac{ub}{1!} + \frac{(ub)^2}{2!} + \frac{(ub)^3}{3!} + \dots$$

$$e^{au} = 1 + \frac{ua}{1!} + \frac{(ua)^2}{2!} + \frac{(ua)^3}{3!} + \dots$$

$$e^{bu} - e^{au} = u(b-a) + \frac{u^2(b^2-a^2)}{2!} + \frac{u^3(b^3-a^3)}{3!} + \dots$$

$$\frac{e^{bu} - e^{au}}{u(b-a)} = 1 + \frac{u(a+b)}{2!} + \frac{u^2(b^2+a^2+ab)}{3!} + \dots$$

$$M_X(u) = 1 + \frac{u(a+b)}{2!} + \frac{u^2(b^2+a^2+ab)}{3!} + \dots$$

$$E(X) = \left. \left[\frac{a+b}{2!} + 2u \frac{(b^2+a^2+ab)}{3!} \right] \right|_{u=0}$$

$$E(X) = \frac{a+b}{2}$$

$$E(x^2) = \left. \frac{d^2 M_x(u)}{du^2} \right|_{u=0}$$

$$E(x^2) = \left. \left[\frac{2(b^2 + a^2 + ab)}{3!} + \frac{3! \mu(b^4 - a^4)}{4!(b-a)} \right] \right|_{u=0}$$

$$E(x^2) = \frac{b^2 + a^2 + ab}{3}$$

$$E(x^3) = \left. \frac{d^3 M_x(u)}{du^3} \right|_{u=0}$$

$$E(x^3) = \left. \left[\frac{3!(b^4 - a^4)}{4!(b-a)} + \frac{4b\mu(b^5 - a^5)}{5!(b-a)} \right] \right|_{u=0}$$

$$E(x^3) = \frac{(b^2 + a^2)(b+a)}{4}$$

Q7 (a) If x and y independent $E(xy) = E(x)E(y)$
also

$$\begin{aligned} \text{Cov}_{xy} &= E(xy) - E(x)E(y) \\ \boxed{\text{Cov}_{xy} &= 0} \end{aligned}$$

(b) $y = ax + b$

$$E(y) = a E(x) + b$$

$$E(yx) = E((ax+b)x) = E(ax^2 + bx)$$

$$E(yx) = a E(x^2) + b E(x)$$

$$\text{Cov}_{xy} = E(xy) - E(y)E(x)$$

$$= a E(x^2) + b E(x) - a(E(x))^2 - b E(x)$$

$$\boxed{\text{Cov}_{xy} = a \sigma_x^2}$$

Q8

$$f_{xy}(x, y) = \begin{cases} 2 & x+y \leq 1, x > 0, y > 0 \\ 0 & \text{o.w} \end{cases}$$

(i) $\text{Cov}(X, Y)$

$$\begin{aligned} E(XY) &= \int_0^1 \int_0^{1-x} 2xy \, dx \, dy = \int_0^1 \frac{2x(1-x)^2}{2} \, dx \\ &= \int_0^1 (x^3 + x - 2x^2) \, dx = \left(\frac{1}{4} + \frac{1}{2} - \frac{2}{3} \right) \\ &= \frac{3}{4} - \frac{2}{3} \end{aligned}$$

$$E(XY) = \frac{1}{12}$$

$$\begin{aligned} E(X) &= \int_0^1 \int_0^{1-x} 2x \, dx \, dy = \int_0^1 x(1-x) \, dx \\ &= 2 \left(\frac{x^2}{2} - \frac{x^3}{3} \right) \Big|_0^1 \\ &= \frac{1}{3} \end{aligned}$$

$$\begin{aligned} E(Y) &= \int_0^1 \int_0^{1-x} 2y \, dy \, dx = \int_0^1 \frac{(1-x)^2}{3} \, dx = \left| -\frac{(1-x)^3}{3} \right|_0^1 \\ &= \frac{1}{3} \end{aligned}$$

$$\text{Cov}(X, Y) = E(XY) - E(X)E(Y)$$

$$= \frac{1}{12} - \frac{1}{3} \cdot \frac{1}{3} = \frac{1}{12} - \frac{1}{9} = \underline{\underline{-\frac{1}{36}}}$$

(ii) $\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$

$$\sigma_X^2 = E(X^2) - (E(X))^2 \quad \sigma_Y^2 = E(Y^2) - (E(Y))^2$$

$$E(x^2) = \int_0^1 \int_0^{1-x} 2x^2 dx dy = 2 \int_0^1 x^2(1-x) dx$$

$$= 2 \left(\frac{x^3}{3} - \frac{x^4}{4} \right) \Big|_0^1$$

$$= \frac{1}{6}$$

$$E(y^2) = \int_0^1 \int_0^{1-x} 2y^2 dx dy = \frac{2}{3} \int_0^1 (1-x)^3 dx$$

$$= \frac{2}{3} \left| -\frac{(1-x)^4}{4} \right|_0^1$$

$$= \frac{1}{6}$$

As we have,

$$\sigma_{x^2} = E(x^2) - (E(x))^2$$

$$= \frac{1}{6} - \frac{1}{9} = \frac{1}{18}$$

$$\sigma_y^2 = \frac{1}{18}$$

$$\rho = \frac{-\frac{1}{36}}{\frac{1}{18}} = -\frac{1}{2}$$

Q9

$$f_{xy}(x,y) = \begin{cases} cx+1 & x, y \geq 0 \quad x+y < 1 \\ 0 & \text{o/w} \end{cases}$$

(i) Using Property of pdf: $\int_{-\infty}^{\infty} f_x(x) dx = 1$

$$\int_0^1 \int_0^{1-x} (cx+1) dx dy = 1$$

$$\int_0^1 (Cx+1)(1-x) dx = 1$$

$$\int_0^1 (Cx - Cx^2 + 1 - x) dx = \frac{C}{2} - \frac{C}{3} + 1 - \frac{1}{2} = 1$$

$$\frac{C}{6} = \frac{1}{2}$$

$$\boxed{C=3}$$

$$(ii) \quad f_x(x) = \int_{-\infty}^{\infty} f_{xy}(x,y) dy$$

$$= \int_0^{1-x} (3x+1) dy = \underline{\underline{(3x+1)(1-x)}}$$

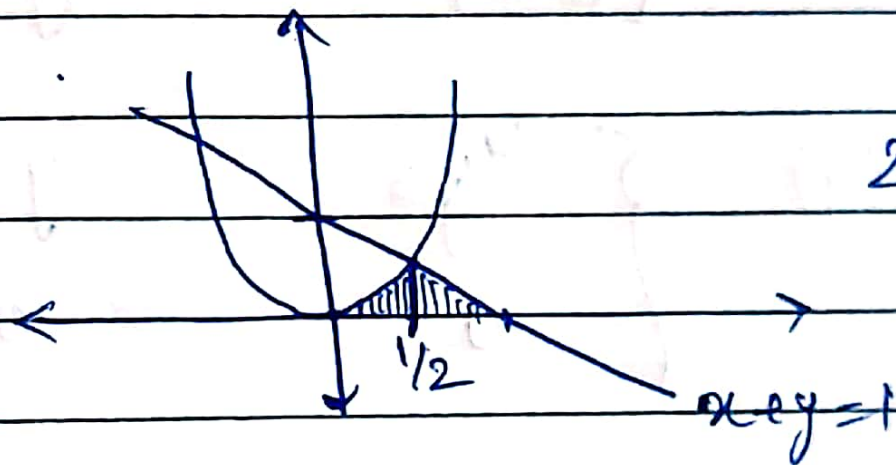
$$f_Y(y) = \int_0^{1-y} (3x+1) dx$$

$$= \left[\frac{3}{2} x^2 + x \right]_0^{1-y} = \frac{3}{2} (1-y)^2 + (1-y)$$

$$f_Y(y) = \frac{1}{2} (1-y) (5-3y)$$

(iii)

$$P[Y < 2x^2]$$



$$x + 2x^2 = 1$$

$$2x^2 + x - 1 = 0$$

$$x = \frac{-1 \pm \sqrt{1+8}}{2 \times 2}$$

$$x = -1, 1/2$$

$$P(4 < 2x^2) = \int_0^{1/2} \int_0^{2x^2} (3x+1) dy dx + \int_{1/2}^1 \int_0^{1-x} (3x+1) dy dx$$

$$= \int_0^{1/2} (3x+1) 2x^2 dx + \int_{1/2}^1 (3x+1)(1-x) dx$$

$$= \int_0^{1/2} (6x^3 + 2x^2) dx + \int_{1/2}^1 (-3x^2 + 2x + 1) dx$$

$$= 6 \left[\frac{1}{2^4} \times \frac{1}{4} + \frac{2}{3} \frac{1}{2^3} \right] + \left[-x^3 + x^2 + x \right]_{1/2}^1$$

$$= \frac{3}{32} + \frac{1}{12} + \left(1 - \left(-\frac{1}{8} + \frac{1}{4} + \frac{1}{2} \right) \right)$$

$$= \frac{3}{32} + \frac{1}{12} + \frac{3}{8}$$

$$= \underline{\underline{0.552}}$$

(10) $f_X(x) = \begin{cases} 4x^2 & 0 < x \leq 1 \\ 0 & \text{o.w} \end{cases}$

$$y = \frac{1}{x}$$

$$\frac{dy}{dx} = -\frac{1}{x^2} \Rightarrow \frac{dy}{dx} = -y^2$$

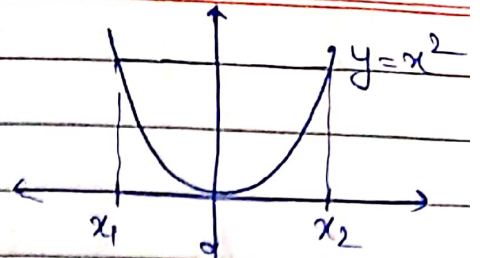
$$f_Y(y) = \frac{f_X(y)}{\left| \frac{dy}{dx} \right|} = \frac{4\left(\frac{1}{y^2}\right)}{\left(\frac{1}{y^2}\right)^2} = \frac{4}{y^2}$$

$$f_Y(y) = \begin{cases} 4/y^2 & y \geq 1 \\ 0 & \text{o.w} \end{cases}$$

(11)

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} \quad \forall x$$

$$y = x^2$$



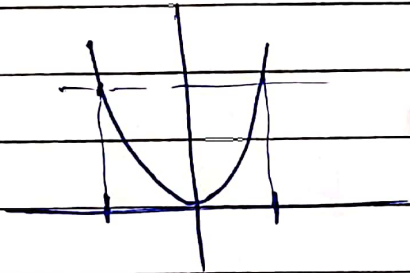
$$f_Y(y) = \frac{f_X(-\sqrt{y})}{\left| \frac{dy}{dx} \right|_{x=x_1}} + \frac{f_X(\sqrt{y})}{\left| \frac{dy}{dx} \right|_{x=x_2}}$$

$$= \frac{1}{\sqrt{2\pi}} \frac{e^{-y/2}}{|-2\sqrt{y}|} + \frac{1}{\sqrt{2\pi}} \frac{e^{-y/2}}{|2\sqrt{y}|}$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi y}} e^{-y/2}$$

(12)

$$f_X(x) = \frac{1}{2} e^{-|x|} \quad y = x^2$$



$$\begin{aligned} F_Y(y) &= P(Y \leq y) \\ &= P(x^2 \leq y) \\ &= P(-\sqrt{y} \leq x \leq \sqrt{y}) \end{aligned}$$

$$= \int_{-\sqrt{y}}^{\sqrt{y}} \frac{1}{2} e^{-|x|} dx$$

$$= \frac{1}{2} x^2 \int_{-\sqrt{y}}^{\sqrt{y}} e^{-x} dx = (1 - e^{-\sqrt{y}})$$

$$F_Y(y) = \begin{cases} 1 - e^{-\sqrt{y}} & y \geq 0 \\ 0 & o.w \end{cases}$$

(13)

$$f_X(x) = \begin{cases} 4x^3 & 0 < x \leq 1 \\ 0 & \text{o.w} \end{cases}$$

$$P(X \leq 2/3 \mid X > 1/3) = ?$$

$$P(X \leq 2/3 \mid X > 1/3) = \frac{P(1/3 < X \leq 2/3)}{P(X > 1/3)}$$

$$= \frac{\int_{1/3}^{2/3} 4x^3 dx}{1 - \int_0^{1/3} 4x^3 dx} = \frac{\int_{1/3}^{2/3} 4x^3 dx}{\int_{1/3}^1 4x^3 dx}$$

$$= \frac{\left(\frac{2}{3}\right)^4 - \left(\frac{1}{3}\right)^4}{1 - \left(\frac{1}{3}\right)^4} = \frac{15 \cdot 3}{16}$$

$$= \frac{3}{16} \text{ Ans}$$